

BACKFEED COGNITIVE AUTOMATION SERVICE

Product Capabilities Briefing & Technical Specifications



Platform Overview

Backfeed is an intelligent agentic middleware layer designed to sit atop B2B ERP distribution systems (like Eclipse). It automates manual entry, optimizes freight consolidation thresholds, checks credit holds, parses incoming client RFQs, and runs warehouse wire-spooling calculations.

Platform Capabilities Available Today

1. Automated RFQ & BOM Parsing

- *Instant BOM Extraction*: Cleans unstructured distributor RFQs, removing formatting noise, and extracts structured items, quantities, and terms using the Gemini API.
- *Automatic Line Card Matcher*: Heuristically maps generic items to active represented manufacturer brands.

2. Freight & Logistics Optimization

- *Prepaid Freight Upsell Detector*: Automatically tracks order values against individual manufacturer freight thresholds and recommends items to add to meet free shipping requirements.
- *Smart-Cut & Strut Simulator*: Performs weight, volume, spool type, and CNC cut-time calculations to eliminate guessing on custom logistics quotes.

3. Sentinel Threat Protection & MS Teams Integration

- *Malicious URL Guard*: Continuously checks IMAP emails using LLM classification, quarantine-deletes threats without navigating URLs, and caches targets in local threat databases.
- *Teams Webhook Alerting*: Posts urgent alerts (such as pricing surges or threat blocks) to your MS Teams feeds headlessly.

4. Environment & Deployment Tooling

- *Unified 1-Click Installer*: Interactive script supporting native Windows setups, automatic port release processes, and container orchestration.
- *Setup Wizard & Path Portability*: CLI wizard dynamically configures runtime variables, automatically remapping directories when run inside containers.

5. Demo Scenario Engine

- *B2B Demonstration Suite*: Preloaded with 12 simulation pathways spanning credit hold overrides, emergency weather rerouting, and active phishing campaigns.

Deployment & Installation Guide

You can spin up the Backfeed platform using either Docker containerization or natively on a Windows Host via PowerShell:

Option A: Docker Containerization (Recommended)

1. Verify that **Docker Desktop** is running on the host system.
2. Launch **install.py** and select **Option [2]** (or execute `docker-compose up --build -d` in a terminal).
3. The container maps the workspace for live hot-reloads on port 8085, bypassing hardware locks using headless activation.

Option B: PowerShell Host Setup (Native Windows)

1. Ensure Python 3.10+ is active and registered on the system environment PATH.
2. Run **install.py** and select **Option [1]** (Native Installation) to register the pywin32 Windows COM automation hooks, install package requirements, and start the local server/background listener.
3. Select **Option [3]** to run the environment credentials modifier wizard, creating your local `.env` credentials.